

Population Structure

John D. Robinson

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Characterizing Genetic Structure across Georgia’s Black Bear Populations

In this markdown we’ll use some files created in **diversity** to characterize population differentiation across Georgia. Two primary analyses are presented below:

1. Calculation of pairwise F_{ST}
2. Clustering analyses (PCA and **admixture**)

But first, to avoid including identical genotypes in the pairwise F_{ST} calculations and the **admixture** analyses below, we will first drop the samples that were duplicated in **combined.vcf** - the “survivors”, bears that were detected in both 2012 and 2023. We wanted to include those samples in both populations for diversity analyses, but here we’ll only keep them in the 2012 group.

Deal with “survivors”

```
CGA12 <- read.table("CGA12keep.txt", header = FALSE)
dim(CGA12)
```

```
## [1] 15  1
```

```
CGA23 <- read.table("CGA23keep.txt", header = FALSE)
dim(CGA23)
```

```
## [1] 169  1
```

```
CGA23nodup <- CGA23$V1[-which(CGA23$V1 %in% CGA12$V1)]
length(CGA23nodup)
```

```
## [1] 166
```

```
write.table(CGA23nodup, file = "CGA23nodup.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
```

Now extract the 2023 samples only, this way we don't have duplicated genotypes in our analysis.

```
#Output the CGA23 only samples
vcftools --vcf ../snppfiltering/PopGen.vcf --keep CGA23nodup.txt \
        --recode --out CGA23nodup

#Grab header and SNP rows
grep -v ^## CGA23nodup.recode.vcf | cut -f 10- > CGA23nodupsnps

#Paste data columns together (using objects made in diversity)
paste CGA12snps CGA23nodupsnps NGAsnps SGAsnps > tempstruct

#Grab header from one of the files
grep ^## CGA12.recode.vcf > structure.vcf
cat tempstruct >> structure.vcf

grep -v ^## structure.vcf | head -1 | tr '\t' '\n' | grep ^UGA > structIID.txt
```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snppfiltering/PopGen.vcf
--keep CGA23nodup.txt
--out CGA23nodup
--recode
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
```

Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 166 out of 260 Individuals
Outputting VCF file...
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 1.00 seconds

Pairwise F_{ST} in PLINK

For this analysis, we first need to make a population “phenotype” file for PLINK2...

```
#Load the metadata file
samps <- read.csv("../input/sampleMetaData_cloneID.csv", header = TRUE)

#Use the IID object created above to build plinkPops
IID <- read.table("structIID.txt", header = FALSE)
FID <- c()
for(i in 1:nrow(IID)) {
  if(strsplit(IID[i,1],split="_")[[1]][1] == "UGAh") {
    FID[i] <- samps$region[which(samps$vcfID == gsub("h","",IID[i,1]))]
  } else {
    FID[i] <- samps$region[which(samps$vcfID == IID[i,1])]
  }

  if(FID[i] %in% c("CGA", "CGA-H")) {
    if(strsplit(IID[i,1],split="_")[[1]][1] == "UGAh") {
      FID[i] <- "CGA12"
    } else {
      FID[i] <- "CGA23"
    }
  }
}
table(FID)
```

```
## FID
## CGA12 CGA23   NGA   SGA
##    15   166    31    26
```

```
plinkPops <- data.frame(FID=FID, IID=IID, pop=FID)

#Write this as a phenotype file
write.table(plinkPops, file = "plinkPops.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
system("echo '#FID IID pop' > plinkSampInfo.txt")
system("cat plinkPops.txt >> plinkSampInfo.txt")
```

Now use the `--fst` argument in PLINK2 to calculate two measures of F_{ST} :

1. [Weir & Cockerham \(1984\)](#) - 'method=wc'

2. The Hudson estimator ([Hudson et al. 1992](#), [Bhatia et al. 2013](#)) - 'method=hudson'

```
plink2 --vcf structure.vcf --fam plinkSampInfo.txt \
      --fst pop 'method=wc' --out WC --silent

plink2 --vcf structure.vcf --fam plinkSampInfo.txt \
      --fst pop 'method=hudson' --out HUDSON --silent
```

Read the output into R...

```
read.table("WC.fst.summary", header = FALSE)
```

```
##      V1      V2      V3
## 1 CGA12 CGA23 -0.000322734
## 2 CGA12  NGA  0.231188000
## 3 CGA12  SGA  0.241792000
## 4 CGA23  NGA  0.263900000
## 5 CGA23  SGA  0.280384000
## 6  NGA   SGA  0.109340000
```

```
read.table("HUDSON.fst.summary", header = FALSE)
```

```
##      V1      V2      V3
## 1 CGA12 CGA23 0.000249955
## 2 CGA12  NGA  0.248668000
## 3 CGA12  SGA  0.258758000
## 4 CGA23  NGA  0.239451000
## 5 CGA23  SGA  0.251607000
## 6  NGA   SGA  0.109836000
```

We see more genetic differentiation between central GA and north / south GA than between north and south GA, despite those two populations being described as separate subspecies (*Ursus americanus americanus* in the north and *U. a. floridanus* in the south). This pattern matches that reported with microsatellite loci in [Hooker et al. \(2019\)](#).

Clustering analyses

Model-free clustering (PCA)

First using the `adegenet::glPca()` function to visualize clustering across the first three principal components.

```
library(vcfR)
```

```
##
##      *****      ***  vcfR      ***      *****
##      This is vcfR 1.15.0
##      browseVignettes('vcfR') # Documentation
##      citation('vcfR') # Citation
##      *****      *****      *****      *****
```

```
library(ade4)
```

```
## Loading required package: ade4
```

```
##  
##    /// adegenet 2.1.11 is loaded ///////////  
##  
##    > overview: '?adegenet'  
##    > tutorials/doc/questions: 'adegenetWeb()'  
##    > bug reports/feature requests: adegenetIssues()
```

```
library(scales)
```

```
vcf <- read.vcfR("structure.vcf", verbose=TRUE)
```

```
## Scanning file to determine attributes.  
## File attributes:  
##   meta lines: 664  
##   header_line: 665  
##   variant count: 10717  
##   column count: 247  
## Meta line 664 read in.  
## All meta lines processed.  
## gt matrix initialized.  
## Character matrix gt created.  
##   Character matrix gt rows: 10717  
##   Character matrix gt cols: 247  
##   skip: 0  
##   nrows: 10717  
##   row_num: 0  
## Processed variant 1000Processed variant 2000Processed variant 3000Processed variant 4000Processed variant 5000  
## All variants processed
```

```
my_genind <- vcfR2genind(vcf, pop=plinkPops$pop)
```

```
my.pca <- glPca(as.genlight(my_genind@tab), nf = 3)
```

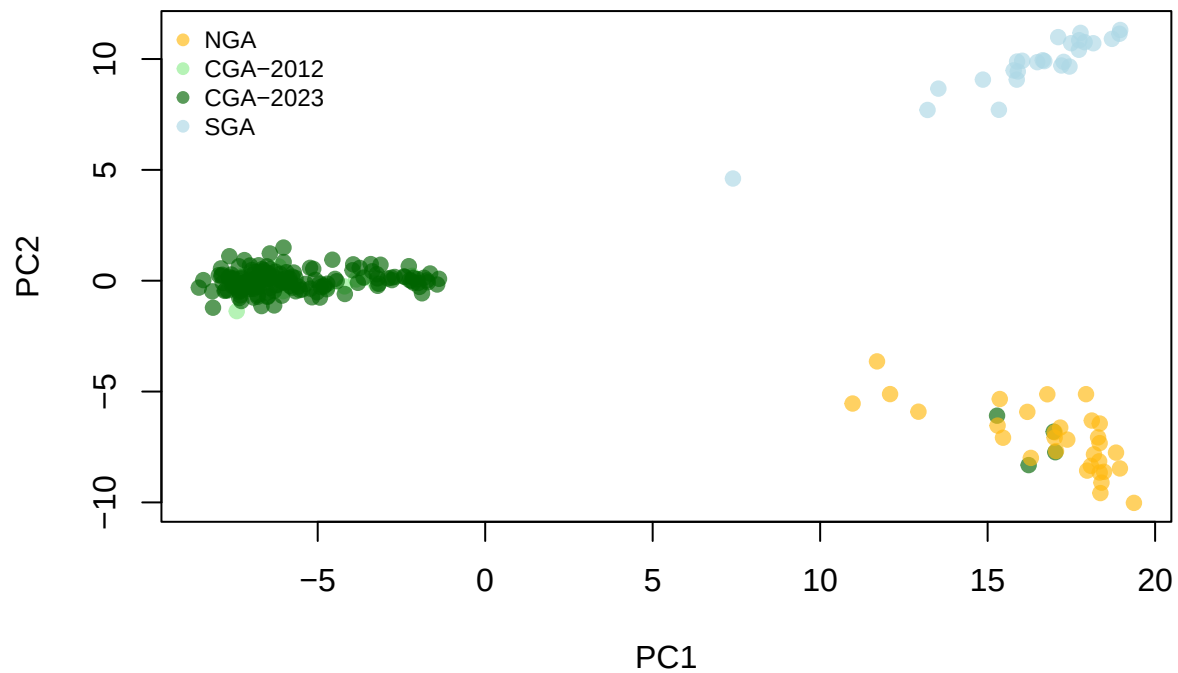
```
transparent.cols <- c(alpha("darkgoldenrod1", 0.65),  
                      alpha("lightgreen", 0.65),  
                      alpha("darkgreen", 0.65),  
                      alpha("lightblue", 0.65))
```

```
ptcolors <- rep("black", length(my_genind@pop))  
ptcolors[my_genind@pop == "NGA"] <- transparent.cols[1]  
ptcolors[my_genind@pop == "CGA12"] <- transparent.cols[2]  
ptcolors[my_genind@pop == "CGA23"] <- transparent.cols[3]  
ptcolors[my_genind@pop == "SGA"] <- transparent.cols[4]
```

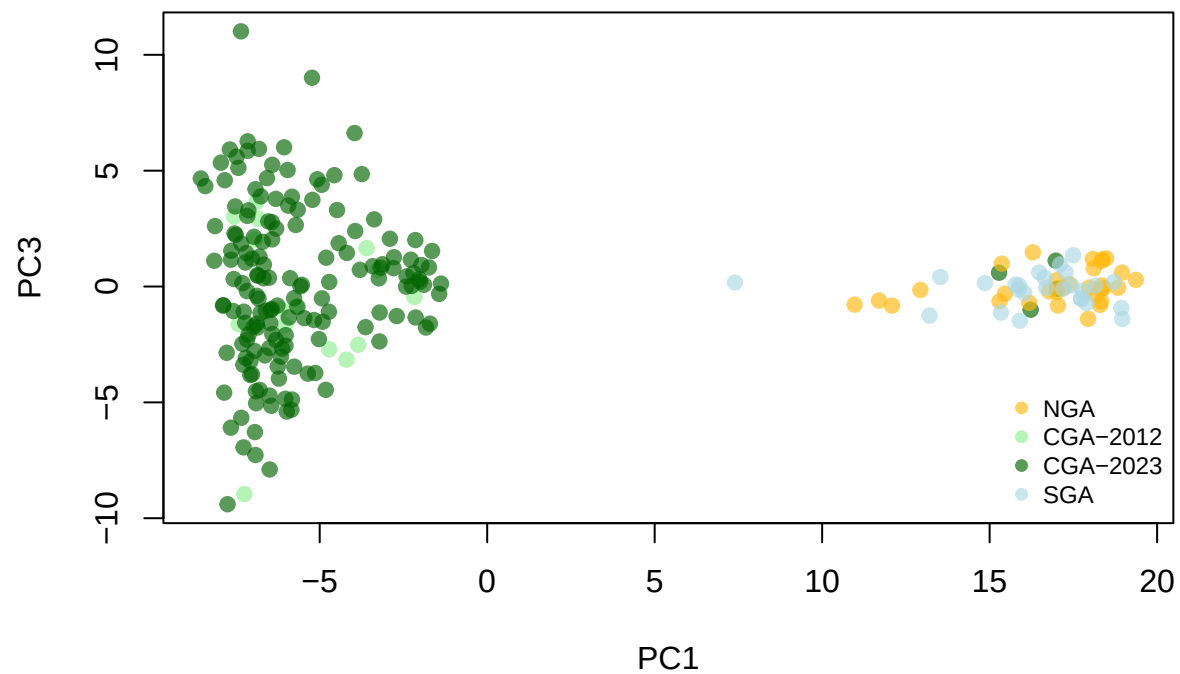
```
table(ptcolors)
```

```
## ptcolors  
## #006400A6 #90EE90A6 #ADD8E6A6 #FFB90FA6  
##      166      15      26      31
```

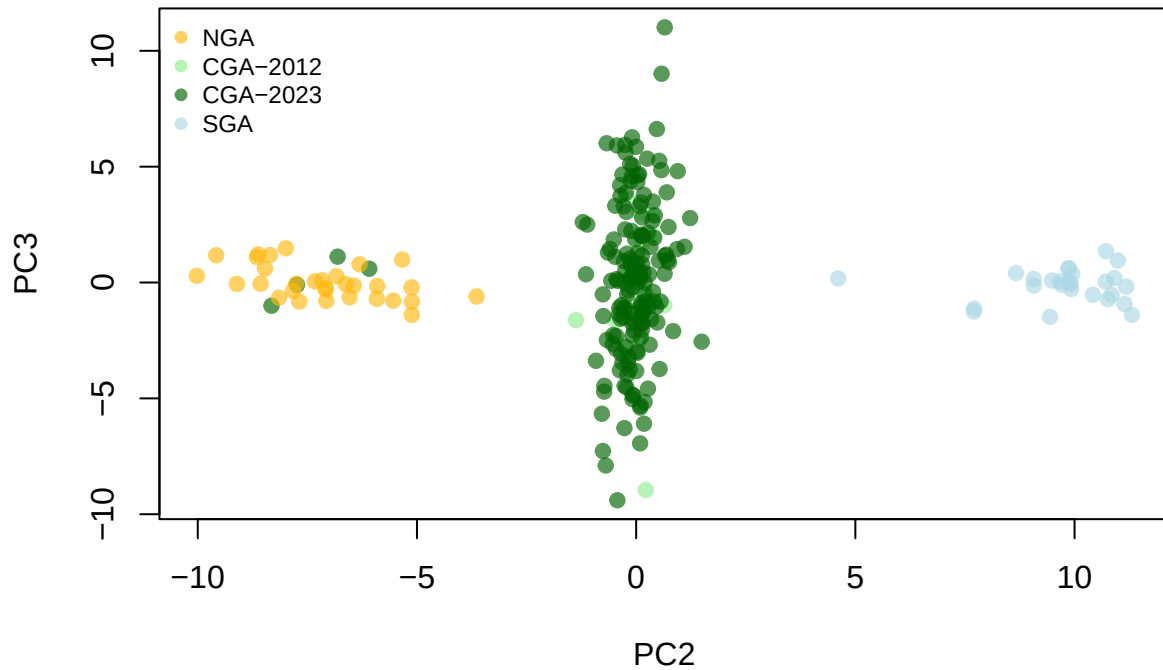
```
#PC1vPC2
plot(my.pca$scores[,1], my.pca$scores[,2], pch = 19, col = ptcolors,
     xlab = "PC1", ylab = "PC2")
legend("topleft", bty = "n", pch = c(19,19,19),
     col=transparent.cols,
     legend = c("NGA", "CGA-2012", "CGA-2023", "SGA"), cex = 0.75)
```



```
#PC1vPC3
plot(my.pca$scores[,1], my.pca$scores[,3], pch = 19, col = ptcolors,
     xlab = "PC1", ylab = "PC3")
legend("bottomright", bty = "n", pch = c(19,19,19),
     col=transparent.cols,
     legend = c("NGA", "CGA-2012", "CGA-2023", "SGA"), cex = 0.75)
```



```
#PC2vPC3
plot(my.pca$scores[,2], my.pca$scores[,3], pch = 19, col = ptc colors,
      xlab = "PC2", ylab = "PC3")
legend("topleft", bty = "n", pch = c(19,19,19),
      col=transparent.cols,
      legend = c("NGA","CGA-2012","CGA-2023","SGA"), cex = 0.75)
```



There are four samples from CGA that are clustering with samples from NGA...

```
#Identify CGA samples clustering with NGA...
```

```
Qsamps <- which(my.pca$scores[,1] > 0 & ptcors == transparent.cols[3])
samps[which(samps$vcfID %in% names(Qsamps)),]
```

```
##               extID ugaID is.SCR is.rep week year snare utmSnare
## 21  UGA_173101_P015_WB09 74-B FALSE FALSE <NA> 2023 <NA>    <NA>
## 31  UGA_173101_P015_WC07 73-B FALSE FALSE <NA> 2023 <NA>    <NA>
## 153 UGA_173101_P016_WE09 83-RK FALSE FALSE <NA> 2023 <NA>    <NA>
## 350 UGA_173101_P018_WF02 89-RK FALSE FALSE <NA> 2024 <NA>    <NA>
##
##                                     fqfile
## 21  RAPiD-Genomics_F534_UGA_173101_P015_WB09_i5-502_i7-63_S14759_L003_R1_001.fastq
## 31  RAPiD-Genomics_F534_UGA_173101_P015_WC07_i5-502_i7-62_S14769_L003_R1_001.fastq
## 153 RAPiD-Genomics_F534_UGA_173101_P016_WE09_i5-504_i7-153_S14890_L003_R1_001.fastq
## 350 RAPiD-Genomics_F534_UGA_173101_P018_WF02_i5-537_i7-158_S15089_L003_R1_001.fastq
##      knownSEX sampSOURCE region nSNPgt          vcfID  FILT GTservice
## 21          M   Den Check   CGA   1234 UGA_173101_P015_WB09 FALSE      LGC
## 31          F   Den Check   CGA   1126 UGA_173101_P015_WC07 FALSE      LGC
## 153         M   Roadkill   CGA   1061 UGA_173101_P016_WE09 FALSE      LGC
## 350         M   Roadkill   CGA   1244 UGA_173101_P018_WF02 FALSE      LGC
##      sexGT COLclone finalClone finalSEX
## 21      M      86      86      M
## 31      F     136     136      F
## 153     M     155     155      M
## 350     M      68      68      M
```


These samples are actually offspring from a NGA sow that was relocated to CGA. She moved north to Hancock / Greene Co. and mated with a male in the area. These samples were either collected from the cubs in the field during a den check (73-B and 74-B), or collected from bears that were hit by motorists (83-RK and 89-RK). It makes sense that they're clustering with NGA bears, but they are not natural migrants from NGA.

Model-based clustering (admixture)

Next we'll use the model-based clustering analysis in `admixture` ([Alexander et al. 2009](#)) to infer group membership. This is the same underlying model used by `STRUCTURE`, but the maximum likelihood implementation is much faster.

We can use [PLINK v. 1.9](#) to create a `.bed` file input for `admixture`, still working from the `structure.vcf` generated above.

```
plink --vcf structure.vcf --recode12 --out ADMIX \
      --double-id --allow-extra-chr --silent
```

We can download `admixture` binaries (for Mac and Linux) from [David Alexander's GitHub site](#).

Now run `admixture` with `K=1-6...`

```
ADMIX="/home/jdrobins/admixture_linux-1.3.0/admixture"

for K in 1 2 3 4 5 6
do $ADMIX --cv=10 ADMIX.ped $K | tee log${K}.out
done

#Best K choice will give us lowest CV error
grep -h CV log*.out
```

```
**** ADMIXTURE Version 1.3.0 ****
**** Copyright 2008-2015 ****
**** David Alexander, Suyash Shringarpure, ****
**** John Novembre, Ken Lange ****
**** ****
**** Please cite our paper! ****
**** Information at www.genetics.ucla.edu/software/admixture ****
```

```
Cross-validation will be performed. Folds=10.
Random seed: 43
Point estimation method: Block relaxation algorithm
Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
Point estimation will terminate when objective function delta < 0.0001
Estimation of standard errors disabled; will compute point estimates only.
Size of G: 238x10717
Performing five EM steps to prime main algorithm
1 (EM) Elapsed: 0.045 Loglikelihood: -721423 (delta): 3.36698e+06
2 (EM) Elapsed: 0.046 Loglikelihood: -721423 (delta): 0
3 (EM) Elapsed: 0.044 Loglikelihood: -721423 (delta): 0
4 (EM) Elapsed: 0.046 Loglikelihood: -721423 (delta): 0
5 (EM) Elapsed: 0.043 Loglikelihood: -721423 (delta): 0
Initial loglikelihood: -721423
```

```

Starting main algorithm
1 (QN/Block)   Elapsed: 0.124   Loglikelihood: -721423   (delta): 0
Summary:
Converged in 1 iterations (0.485 sec)
Loglikelihood: -721422.978684
CV error (K=1): 0.23947
Writing output files.
****              ADMIXTURE Version 1.3.0              ****
****              Copyright 2008-2015                  ****
****              David Alexander, Suyash Shringarpure, ****
****              John Novembre, Ken Lange              ****
****              Please cite our paper!                ****
****              Information at www.genetics.ucla.edu/software/admixture ****

Cross-validation will be performed.   Folds=10.
Random seed: 43
Point estimation method: Block relaxation algorithm
Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
Point estimation will terminate when objective function delta < 0.0001
Estimation of standard errors disabled; will compute point estimates only.
Size of G: 238x10717
Performing five EM steps to prime main algorithm
1 (EM) Elapsed: 0.074   Loglikelihood: -753122   (delta): 2.77048e+06
2 (EM) Elapsed: 0.074   Loglikelihood: -727001   (delta): 26120.9
3 (EM) Elapsed: 0.074   Loglikelihood: -720961   (delta): 6039.91
4 (EM) Elapsed: 0.074   Loglikelihood: -718345   (delta): 2615.97
5 (EM) Elapsed: 0.074   Loglikelihood: -717173   (delta): 1172.4
Initial loglikelihood: -717173
Starting main algorithm
1 (QN/Block)   Elapsed: 0.178   Loglikelihood: -669728   (delta): 47444.5
2 (QN/Block)   Elapsed: 0.177   Loglikelihood: -645226   (delta): 24502.7
3 (QN/Block)   Elapsed: 0.212   Loglikelihood: -636745   (delta): 8480.87
4 (QN/Block)   Elapsed: 0.259   Loglikelihood: -625441   (delta): 11304.1
5 (QN/Block)   Elapsed: 0.242   Loglikelihood: -620138   (delta): 5302.76
6 (QN/Block)   Elapsed: 0.258   Loglikelihood: -618905   (delta): 1233.02
7 (QN/Block)   Elapsed: 0.21    Loglikelihood: -618439   (delta): 465.685
8 (QN/Block)   Elapsed: 0.258   Loglikelihood: -618214   (delta): 225.578
9 (QN/Block)   Elapsed: 0.21    Loglikelihood: -618123   (delta): 90.5119
10 (QN/Block)  Elapsed: 0.21    Loglikelihood: -618082   (delta): 40.8223
11 (QN/Block)  Elapsed: 0.258   Loglikelihood: -618060   (delta): 22.1425
12 (QN/Block)  Elapsed: 0.226   Loglikelihood: -618054   (delta): 6.00732
13 (QN/Block)  Elapsed: 0.257   Loglikelihood: -618054   (delta): 0.01496
14 (QN/Block)  Elapsed: 0.21    Loglikelihood: -618054   (delta): 8.11461e-06
Summary:
Converged in 14 iterations (3.894 sec)
Loglikelihood: -618054.234607
Fst divergences between estimated populations:
  Pop0
  Pop1    0.208
CV error (K=2): 0.20200
Writing output files.
****              ADMIXTURE Version 1.3.0              ****

```

```

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****                      Information at www.genetics.ucla.edu/software/admixture ****

Cross-validation will be performed.  Folds=10.
Random seed: 43
Point estimation method: Block relaxation algorithm
Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
Point estimation will terminate when objective function delta < 0.0001
Estimation of standard errors disabled; will compute point estimates only.
Size of G: 238x10717
Performing five EM steps to prime main algorithm
1 (EM) Elapsed: 0.101 Loglikelihood: -768414 (delta): 2.5143e+06
2 (EM) Elapsed: 0.101 Loglikelihood: -729770 (delta): 38643.3
3 (EM) Elapsed: 0.101 Loglikelihood: -721743 (delta): 8027.41
4 (EM) Elapsed: 0.101 Loglikelihood: -717238 (delta): 4504.79
5 (EM) Elapsed: 0.101 Loglikelihood: -714373 (delta): 2865.25
Initial loglikelihood: -714373
Starting main algorithm
1 (QN/Block) Elapsed: 0.245 Loglikelihood: -656428 (delta): 57944.9
2 (QN/Block) Elapsed: 0.245 Loglikelihood: -625507 (delta): 30920.7
3 (QN/Block) Elapsed: 0.283 Loglikelihood: -613492 (delta): 12014.9
4 (QN/Block) Elapsed: 0.285 Loglikelihood: -604198 (delta): 9294.49
5 (QN/Block) Elapsed: 0.302 Loglikelihood: -600676 (delta): 3521.65
6 (QN/Block) Elapsed: 0.285 Loglikelihood: -600013 (delta): 662.779
7 (QN/Block) Elapsed: 0.284 Loglikelihood: -599381 (delta): 632.459
8 (QN/Block) Elapsed: 0.335 Loglikelihood: -599136 (delta): 244.869
9 (QN/Block) Elapsed: 0.284 Loglikelihood: -598991 (delta): 144.749
10 (QN/Block) Elapsed: 0.318 Loglikelihood: -598967 (delta): 24.2108
11 (QN/Block) Elapsed: 0.285 Loglikelihood: -598963 (delta): 4.04932
12 (QN/Block) Elapsed: 0.336 Loglikelihood: -598962 (delta): 1.38786
13 (QN/Block) Elapsed: 0.318 Loglikelihood: -598962 (delta): 0.0915383
14 (QN/Block) Elapsed: 0.335 Loglikelihood: -598962 (delta): 3.06832e-05
Summary:
Converged in 14 iterations (5.031 sec)
Loglikelihood: -598961.508477
Fst divergences between estimated populations:
  Pop0    Pop1
Pop0
Pop1    0.118
Pop2    0.230    0.238
CV error (K=3): 0.20075
Writing output files.
****                      ADMIXTURE Version 1.3.0                      ****
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****                      John Novembre, Ken Lange                  ****
****                      Please cite our paper!                    ****
****                      Information at www.genetics.ucla.edu/software/admixture ****

```

Cross-validation will be performed. Folds=10.
Random seed: 43
Point estimation method: Block relaxation algorithm
Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
Point estimation will terminate when objective function delta < 0.0001
Estimation of standard errors disabled; will compute point estimates only.
Size of G: 238x10717
Performing five EM steps to prime main algorithm
1 (EM) Elapsed: 0.13 Loglikelihood: -776040 (delta): 2.40278e+06
2 (EM) Elapsed: 0.13 Loglikelihood: -731069 (delta): 44970.8
3 (EM) Elapsed: 0.13 Loglikelihood: -722393 (delta): 8676.14
4 (EM) Elapsed: 0.13 Loglikelihood: -717197 (delta): 5196.36
5 (EM) Elapsed: 0.13 Loglikelihood: -713516 (delta): 3681.37
Initial loglikelihood: -713516
Starting main algorithm
1 (QN/Block) Elapsed: 0.332 Loglikelihood: -640147 (delta): 73368.4
2 (QN/Block) Elapsed: 0.351 Loglikelihood: -616492 (delta): 23655.1
3 (QN/Block) Elapsed: 0.403 Loglikelihood: -603771 (delta): 12720.9
4 (QN/Block) Elapsed: 0.422 Loglikelihood: -593067 (delta): 10704
5 (QN/Block) Elapsed: 0.44 Loglikelihood: -587915 (delta): 5151.62
6 (QN/Block) Elapsed: 0.406 Loglikelihood: -585688 (delta): 2227.39
7 (QN/Block) Elapsed: 0.404 Loglikelihood: -585169 (delta): 518.982
8 (QN/Block) Elapsed: 0.388 Loglikelihood: -584975 (delta): 194.01
9 (QN/Block) Elapsed: 0.387 Loglikelihood: -584908 (delta): 66.5793
10 (QN/Block) Elapsed: 0.386 Loglikelihood: -584874 (delta): 34.6824
11 (QN/Block) Elapsed: 0.444 Loglikelihood: -584842 (delta): 32.0333
12 (QN/Block) Elapsed: 0.441 Loglikelihood: -584821 (delta): 21.2438
13 (QN/Block) Elapsed: 0.441 Loglikelihood: -584777 (delta): 43.3674
14 (QN/Block) Elapsed: 0.386 Loglikelihood: -584755 (delta): 22.4111
15 (QN/Block) Elapsed: 0.403 Loglikelihood: -584752 (delta): 2.55518
16 (QN/Block) Elapsed: 0.403 Loglikelihood: -584752 (delta): 0.0215868
17 (QN/Block) Elapsed: 0.389 Loglikelihood: -584752 (delta): 0.00440354
18 (QN/Block) Elapsed: 0.388 Loglikelihood: -584752 (delta): 0.000403958
19 (QN/Block) Elapsed: 0.39 Loglikelihood: -584752 (delta): 7.78304e-05
Summary:
Converged in 19 iterations (8.754 sec)
Loglikelihood: -584752.171940
Fst divergences between estimated populations:
Pop0 Pop1 Pop2
Pop0
Pop1 0.242
Pop2 0.249 0.119
Pop3 0.052 0.215 0.217
CV error (K=4): 0.20170
Writing output files.
**** ADMIXTURE Version 1.3.0 ****
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**** John Novembre, Ken Lange ****
**** ****
**** Please cite our paper! ****
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Cross-validation will be performed. Folds=10.

Random seed: 43
 Point estimation method: Block relaxation algorithm
 Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
 Point estimation will terminate when objective function delta < 0.0001
 Estimation of standard errors disabled; will compute point estimates only.
 Size of G: 238x10717
 Performing five EM steps to prime main algorithm
 1 (EM) Elapsed: 0.161 Loglikelihood: -778570 (delta): 2.331e+06
 2 (EM) Elapsed: 0.161 Loglikelihood: -730910 (delta): 47660.6
 3 (EM) Elapsed: 0.16 Loglikelihood: -722829 (delta): 8080.5
 4 (EM) Elapsed: 0.161 Loglikelihood: -718042 (delta): 4787.37
 5 (EM) Elapsed: 0.161 Loglikelihood: -714594 (delta): 3447.46
 Initial loglikelihood: -714594
 Starting main algorithm
 1 (QN/Block) Elapsed: 0.446 Loglikelihood: -637362 (delta): 77232.6
 2 (QN/Block) Elapsed: 0.458 Loglikelihood: -609393 (delta): 27968.8
 3 (QN/Block) Elapsed: 0.488 Loglikelihood: -596369 (delta): 13024.4
 4 (QN/Block) Elapsed: 0.504 Loglikelihood: -587823 (delta): 8545.86
 5 (QN/Block) Elapsed: 0.544 Loglikelihood: -581958 (delta): 5864.46
 6 (QN/Block) Elapsed: 0.547 Loglikelihood: -579050 (delta): 2908.02
 7 (QN/Block) Elapsed: 0.522 Loglikelihood: -577947 (delta): 1103.03
 8 (QN/Block) Elapsed: 0.477 Loglikelihood: -577654 (delta): 293.237
 9 (QN/Block) Elapsed: 0.484 Loglikelihood: -577472 (delta): 182.006
 10 (QN/Block) Elapsed: 0.484 Loglikelihood: -577344 (delta): 127.604
 11 (QN/Block) Elapsed: 0.484 Loglikelihood: -577279 (delta): 65.6524
 12 (QN/Block) Elapsed: 0.483 Loglikelihood: -577250 (delta): 29.027
 13 (QN/Block) Elapsed: 0.484 Loglikelihood: -577238 (delta): 11.1881
 14 (QN/Block) Elapsed: 0.503 Loglikelihood: -577227 (delta): 11.6085
 15 (QN/Block) Elapsed: 0.483 Loglikelihood: -577195 (delta): 31.5686
 16 (QN/Block) Elapsed: 0.484 Loglikelihood: -577129 (delta): 66.3012
 17 (QN/Block) Elapsed: 0.485 Loglikelihood: -577088 (delta): 40.7837
 18 (QN/Block) Elapsed: 0.542 Loglikelihood: -577083 (delta): 4.8333
 19 (QN/Block) Elapsed: 0.504 Loglikelihood: -577082 (delta): 1.43062
 20 (QN/Block) Elapsed: 0.481 Loglikelihood: -577081 (delta): 0.886678
 21 (QN/Block) Elapsed: 0.544 Loglikelihood: -577080 (delta): 1.01047
 22 (QN/Block) Elapsed: 0.551 Loglikelihood: -577079 (delta): 0.57185
 23 (QN/Block) Elapsed: 0.541 Loglikelihood: -577079 (delta): 0.399612
 24 (QN/Block) Elapsed: 0.551 Loglikelihood: -577079 (delta): 0.431241
 25 (QN/Block) Elapsed: 0.542 Loglikelihood: -577078 (delta): 0.359397
 26 (QN/Block) Elapsed: 0.522 Loglikelihood: -577078 (delta): 0.464316
 27 (QN/Block) Elapsed: 0.523 Loglikelihood: -577077 (delta): 0.537884
 28 (QN/Block) Elapsed: 0.503 Loglikelihood: -577077 (delta): 0.324631
 29 (QN/Block) Elapsed: 0.484 Loglikelihood: -577077 (delta): 0.423245
 30 (QN/Block) Elapsed: 0.504 Loglikelihood: -577076 (delta): 0.414335
 31 (QN/Block) Elapsed: 0.543 Loglikelihood: -577075 (delta): 0.764476
 32 (QN/Block) Elapsed: 0.542 Loglikelihood: -577075 (delta): 0.768822
 33 (QN/Block) Elapsed: 0.542 Loglikelihood: -577074 (delta): 0.818197
 34 (QN/Block) Elapsed: 0.543 Loglikelihood: -577073 (delta): 0.989384
 35 (QN/Block) Elapsed: 0.525 Loglikelihood: -577072 (delta): 0.931853
 36 (QN/Block) Elapsed: 0.545 Loglikelihood: -577071 (delta): 0.855863
 37 (QN/Block) Elapsed: 0.503 Loglikelihood: -577070 (delta): 0.627475
 38 (QN/Block) Elapsed: 0.529 Loglikelihood: -577069 (delta): 0.969243
 39 (QN/Block) Elapsed: 0.524 Loglikelihood: -577068 (delta): 1.27167
 40 (QN/Block) Elapsed: 0.504 Loglikelihood: -577068 (delta): 0.59551

41 (QN/Block)	Elapsed: 0.483	Loglikelihood: -577067	(delta): 0.447617
42 (QN/Block)	Elapsed: 0.501	Loglikelihood: -577067	(delta): 0.265613
43 (QN/Block)	Elapsed: 0.544	Loglikelihood: -577067	(delta): 0.276193
44 (QN/Block)	Elapsed: 0.484	Loglikelihood: -577066	(delta): 0.282617
45 (QN/Block)	Elapsed: 0.543	Loglikelihood: -577065	(delta): 1.3789
46 (QN/Block)	Elapsed: 0.543	Loglikelihood: -577062	(delta): 2.47374
47 (QN/Block)	Elapsed: 0.537	Loglikelihood: -577060	(delta): 2.00178
48 (QN/Block)	Elapsed: 0.545	Loglikelihood: -577059	(delta): 0.939734
49 (QN/Block)	Elapsed: 0.486	Loglikelihood: -577059	(delta): 0.619051
50 (QN/Block)	Elapsed: 0.502	Loglikelihood: -577058	(delta): 0.823947
51 (QN/Block)	Elapsed: 0.546	Loglikelihood: -577056	(delta): 1.71489
52 (QN/Block)	Elapsed: 0.513	Loglikelihood: -577055	(delta): 1.50074
53 (QN/Block)	Elapsed: 0.503	Loglikelihood: -577052	(delta): 2.43438
54 (QN/Block)	Elapsed: 0.484	Loglikelihood: -577048	(delta): 3.95531
55 (QN/Block)	Elapsed: 0.483	Loglikelihood: -577037	(delta): 11.7996
56 (QN/Block)	Elapsed: 0.542	Loglikelihood: -577005	(delta): 31.2223
57 (QN/Block)	Elapsed: 0.503	Loglikelihood: -576991	(delta): 14.1898
58 (QN/Block)	Elapsed: 0.491	Loglikelihood: -576988	(delta): 3.21138
59 (QN/Block)	Elapsed: 0.526	Loglikelihood: -576986	(delta): 1.70346
60 (QN/Block)	Elapsed: 0.496	Loglikelihood: -576985	(delta): 0.885226
61 (QN/Block)	Elapsed: 0.543	Loglikelihood: -576985	(delta): 0.667748
62 (QN/Block)	Elapsed: 0.544	Loglikelihood: -576984	(delta): 0.609244
63 (QN/Block)	Elapsed: 0.543	Loglikelihood: -576984	(delta): 0.49966
64 (QN/Block)	Elapsed: 0.523	Loglikelihood: -576983	(delta): 0.284592
65 (QN/Block)	Elapsed: 0.523	Loglikelihood: -576983	(delta): 0.12439
66 (QN/Block)	Elapsed: 0.504	Loglikelihood: -576983	(delta): 0.042767
67 (QN/Block)	Elapsed: 0.484	Loglikelihood: -576983	(delta): 0.0438092
68 (QN/Block)	Elapsed: 0.542	Loglikelihood: -576983	(delta): 0.13356
69 (QN/Block)	Elapsed: 0.538	Loglikelihood: -576980	(delta): 2.53736
70 (QN/Block)	Elapsed: 0.543	Loglikelihood: -576978	(delta): 2.03701
71 (QN/Block)	Elapsed: 0.542	Loglikelihood: -576978	(delta): 0.393357
72 (QN/Block)	Elapsed: 0.543	Loglikelihood: -576978	(delta): 0.173043
73 (QN/Block)	Elapsed: 0.519	Loglikelihood: -576978	(delta): 0.0364022
74 (QN/Block)	Elapsed: 0.478	Loglikelihood: -576978	(delta): 0.00916462
75 (QN/Block)	Elapsed: 0.483	Loglikelihood: -576978	(delta): 0.0029765
76 (QN/Block)	Elapsed: 0.483	Loglikelihood: -576978	(delta): 0.000571108
77 (QN/Block)	Elapsed: 0.479	Loglikelihood: -576978	(delta): 0.000223765
78 (QN/Block)	Elapsed: 0.483	Loglikelihood: -576978	(delta): 0.000109527
79 (QN/Block)	Elapsed: 0.483	Loglikelihood: -576978	(delta): 1.84516e-05

Summary:

Converged in 79 iterations (43.035 sec)

Loglikelihood: -576977.779042

Fst divergences between estimated populations:

	Pop0	Pop1	Pop2	Pop3
Pop0				
Pop1	0.267			
Pop2	0.258	0.070		
Pop3	0.122	0.258	0.248	
Pop4	0.302	0.262	0.262	0.300

CV error (K=5): 0.20006

Writing output files.

**** ADMIXTURE Version 1.3.0 ****
 **** Copyright 2008-2015 ****
 **** David Alexander, Suyash Shringarpure, ****

```

****              John Novembre, Ken Lange              ****
****              ****
****              Please cite our paper!                ****
****  Information at www.genetics.ucla.edu/software/admixture  ****

```

```

Cross-validation will be performed.  Folds=10.
Random seed: 43
Point estimation method: Block relaxation algorithm
Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
Point estimation will terminate when objective function delta < 0.0001
Estimation of standard errors disabled; will compute point estimates only.
Size of G: 238x10717

```

Performing five EM steps to prime main algorithm

```

1 (EM) Elapsed: 0.196 Loglikelihood: -779311 (delta): 2.27248e+06
2 (EM) Elapsed: 0.197 Loglikelihood: -728570 (delta): 50741.2
3 (EM) Elapsed: 0.196 Loglikelihood: -720189 (delta): 8380.1
4 (EM) Elapsed: 0.196 Loglikelihood: -714934 (delta): 5255.42
5 (EM) Elapsed: 0.196 Loglikelihood: -710726 (delta): 4208.4

```

Initial loglikelihood: -710726

Starting main algorithm

```

1 (QN/Block) Elapsed: 0.561 Loglikelihood: -621045 (delta): 89681.1
2 (QN/Block) Elapsed: 0.559 Loglikelihood: -597612 (delta): 23433
3 (QN/Block) Elapsed: 0.628 Loglikelihood: -585451 (delta): 12160.4
4 (QN/Block) Elapsed: 0.63 Loglikelihood: -577558 (delta): 7893.21
5 (QN/Block) Elapsed: 0.635 Loglikelihood: -573821 (delta): 3736.42
6 (QN/Block) Elapsed: 0.606 Loglikelihood: -572192 (delta): 1629.34
7 (QN/Block) Elapsed: 0.605 Loglikelihood: -571546 (delta): 645.935
8 (QN/Block) Elapsed: 0.606 Loglikelihood: -571304 (delta): 242.28
9 (QN/Block) Elapsed: 0.627 Loglikelihood: -571190 (delta): 113.8
10 (QN/Block) Elapsed: 0.628 Loglikelihood: -571143 (delta): 47.4958
11 (QN/Block) Elapsed: 0.628 Loglikelihood: -571085 (delta): 57.7732
12 (QN/Block) Elapsed: 0.606 Loglikelihood: -571034 (delta): 50.9973
13 (QN/Block) Elapsed: 0.628 Loglikelihood: -570983 (delta): 51.2885
14 (QN/Block) Elapsed: 0.628 Loglikelihood: -570944 (delta): 38.6858
15 (QN/Block) Elapsed: 0.628 Loglikelihood: -570923 (delta): 20.5622
16 (QN/Block) Elapsed: 0.605 Loglikelihood: -570916 (delta): 7.55812
17 (QN/Block) Elapsed: 0.627 Loglikelihood: -570904 (delta): 12.1307
18 (QN/Block) Elapsed: 0.648 Loglikelihood: -570896 (delta): 7.92098
19 (QN/Block) Elapsed: 0.607 Loglikelihood: -570893 (delta): 3.15524
20 (QN/Block) Elapsed: 0.606 Loglikelihood: -570890 (delta): 2.06819
21 (QN/Block) Elapsed: 0.628 Loglikelihood: -570890 (delta): 0.722647
22 (QN/Block) Elapsed: 0.605 Loglikelihood: -570890 (delta): 0.0648208
23 (QN/Block) Elapsed: 0.606 Loglikelihood: -570890 (delta): 0.0305645
24 (QN/Block) Elapsed: 0.605 Loglikelihood: -570890 (delta): 0.0118749
25 (QN/Block) Elapsed: 0.671 Loglikelihood: -570890 (delta): 0.0049352
26 (QN/Block) Elapsed: 0.664 Loglikelihood: -570890 (delta): 0.00729572
27 (QN/Block) Elapsed: 0.671 Loglikelihood: -570890 (delta): 0.0130291
28 (QN/Block) Elapsed: 0.649 Loglikelihood: -570890 (delta): 0.0677799
29 (QN/Block) Elapsed: 0.664 Loglikelihood: -570889 (delta): 0.40221
30 (QN/Block) Elapsed: 0.67 Loglikelihood: -570888 (delta): 0.992714
31 (QN/Block) Elapsed: 0.67 Loglikelihood: -570887 (delta): 0.688183
32 (QN/Block) Elapsed: 0.671 Loglikelihood: -570887 (delta): 0.0264843
33 (QN/Block) Elapsed: 0.607 Loglikelihood: -570887 (delta): 0.000708267
34 (QN/Block) Elapsed: 0.627 Loglikelihood: -570887 (delta): 0.0003235

```

```

35 (QN/Block)   Elapsed: 0.605   Loglikelihood: -570887   (delta): 5.1451e-05
Summary:
Converged in 35 iterations (23.837 sec)
Loglikelihood: -570887.451095
Fst divergences between estimated populations:
  Pop0    Pop1    Pop2    Pop3    Pop4
Pop0
Pop1    0.302
Pop2    0.272    0.123
Pop3    0.156    0.260    0.235
Pop4    0.308    0.085    0.084    0.259
Pop5    0.183    0.254    0.226    0.131    0.253
CV error (K=6): 0.20843
Writing output files.
CV error (K=1): 0.23947
CV error (K=2): 0.20200
CV error (K=3): 0.20075
CV error (K=4): 0.20170
CV error (K=5): 0.20006
CV error (K=6): 0.20843

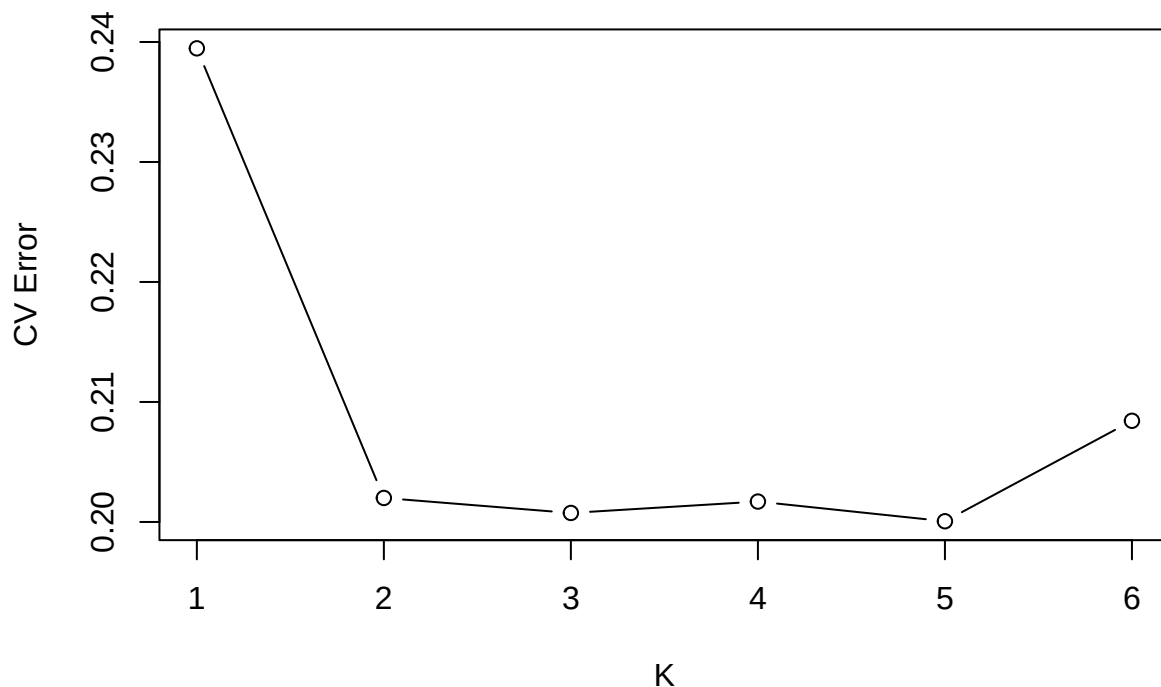
```

Now let's plot the results...

```

## First the CV error plot
CVerr <- system("grep -h CV log*.out | cut -f 2 -d :", intern = TRUE)
CVerr <- as.numeric(CVerr)
plot(x = c(1:6), CVerr, type = "b", xlab = "K", ylab = "CV Error")

```

```
bestK <- which(CVerr == min(CVerr))
```

The best K appears to be $K = 5$, based on our cross-validations. Now let's visualize the results at the different K values...

```
## Now for admixture proportions
#First read in the sample IDs
adsamps <- read.table("ADMIX.nosex", header = FALSE)
popmap <- data.frame(vcfID = adsamps[,1], pop=NA, sortord=NA)
for(i in 1:nrow(popmap)) {
  if(strsplit(popmap$vcfID[i],split="_")[[1]][1] == "UGAh") {
    popmap$pop[i] <- "CGA12"
  } else {
    popmap$pop[i] <- samps$region[which(samps$vcfID == popmap$vcfID[i])]
    if(popmap$pop[i] %in% c("CGA","CGA-H")) {
      popmap$pop[i] <- "CGA23"
    }
  }
}

#Plot results with K values from 2 to 6
sortedQ <- colscheme <- vector("list", 6)
for(k in 2:6) {
  bestKQ <- read.table(paste0("ADMIX.",k,".Q"), header=FALSE)
  dim(bestKQ)
```

```

#Order the color scheme so it lines up with the populations...
popXcol <- data.frame(pop = c("NGA","CGA23","CGA12","SGA"),
                      col = c("darkgoldenrod1","darkgreen",
                              "lightgreen","lightblue"))

colscheme[[k]] <- rep(NA, k)
p <- 1
while(sum(is.na(colscheme[[k]])) > 0) {
  if(p > 4) {
    if(sum(is.na(colscheme[[k]])) == 1) {
      colscheme[[k]][is.na(colscheme[[k]])] <- "lightgrey"
    } else if(sum(is.na(colscheme[[k]])) == 2) {
      colscheme[[k]][is.na(colscheme[[k]])] <- c("lightgrey","darkgrey")
    } else if(sum(is.na(colscheme[[k]])) == 3) {
      colscheme[[k]][is.na(colscheme[[k]])] <- c("lightgrey","darkgrey","black")
    }
  } else {
    tmp <- which(apply(bestKQ[which(popmap$pop == popXcol$pop[p]),],2,mean) > 0.5)
    if(length(tmp) > 0 & is.na(colscheme[[k]][tmp])) {
      colscheme[[k]][tmp] <- popXcol$col[p]
    }
    rm(tmp)
  }
  p <- p+1
}

#Set the order for plotting along the x-axis (individuals)...
popmap$sortord[which(popmap$pop == "NGA")] <- 1
popmap$sortord[which(popmap$pop == "CGA12")] <- 2
popmap$sortord[which(popmap$pop == "CGA23")] <- 3
popmap$sortord[which(popmap$pop == "SGA")] <- 4

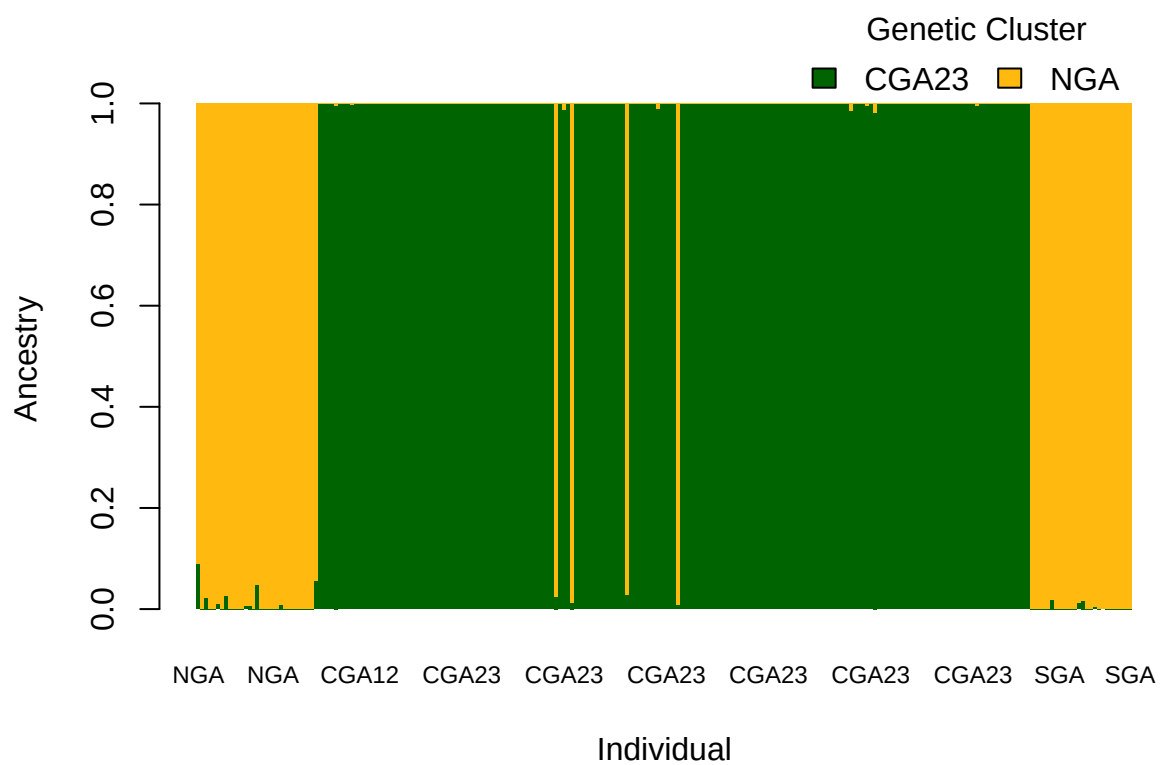
sum(is.na(popmap$sortord))

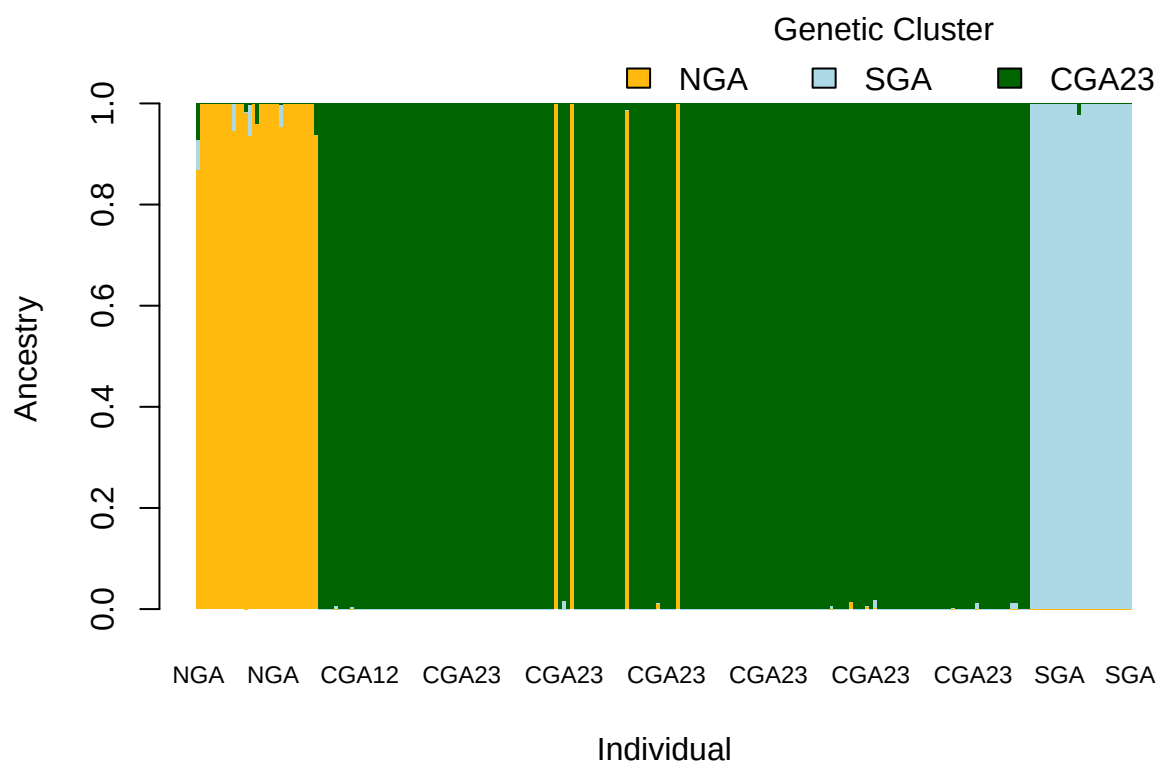
#Sort the samples by population - NGA, CGA12, CGA23, SGA...
sortedmap <- popmap[order(popmap$sortord),]
sortedQ[[k]] <- bestKQ[order(popmap$sortord),]

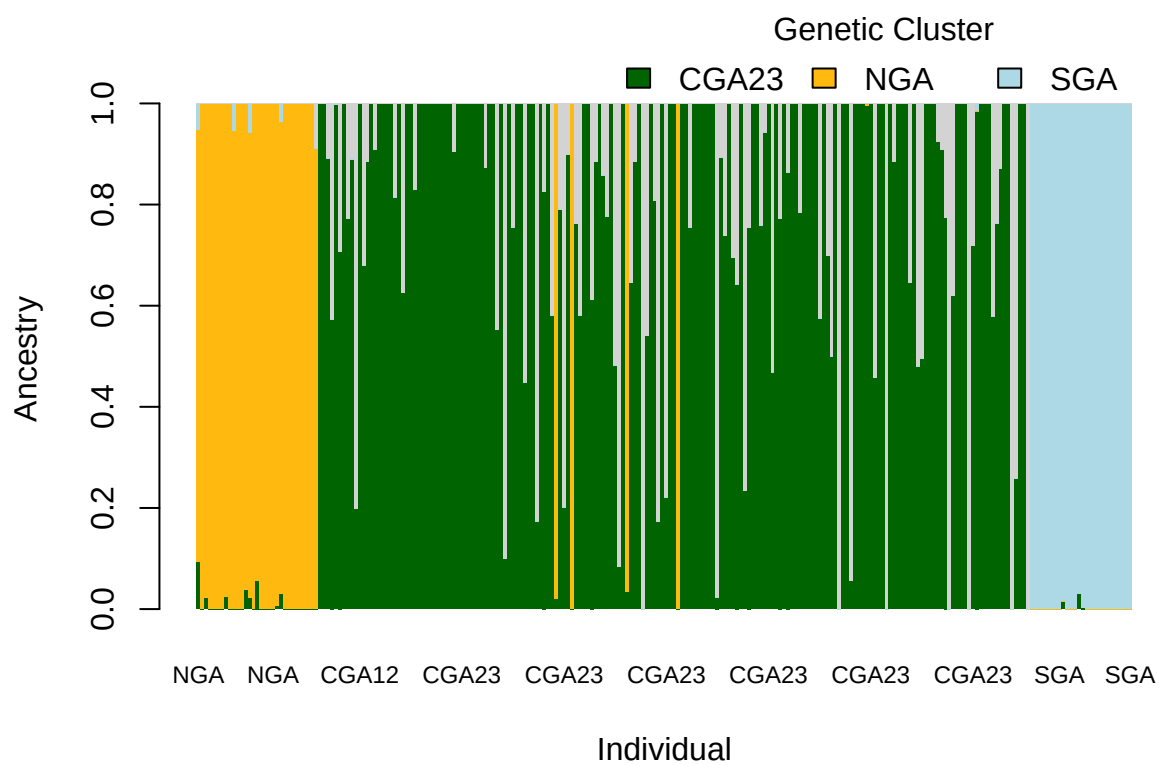
fillcols <- popXcol$col[match(colscheme[[k]], popXcol$col)]
filllleg <- popXcol$pop[match(colscheme[[k]], popXcol$col)]
fillcols <- fillcols[!is.na(fillcols)]
filllleg <- filllleg[!is.na(filllleg)]

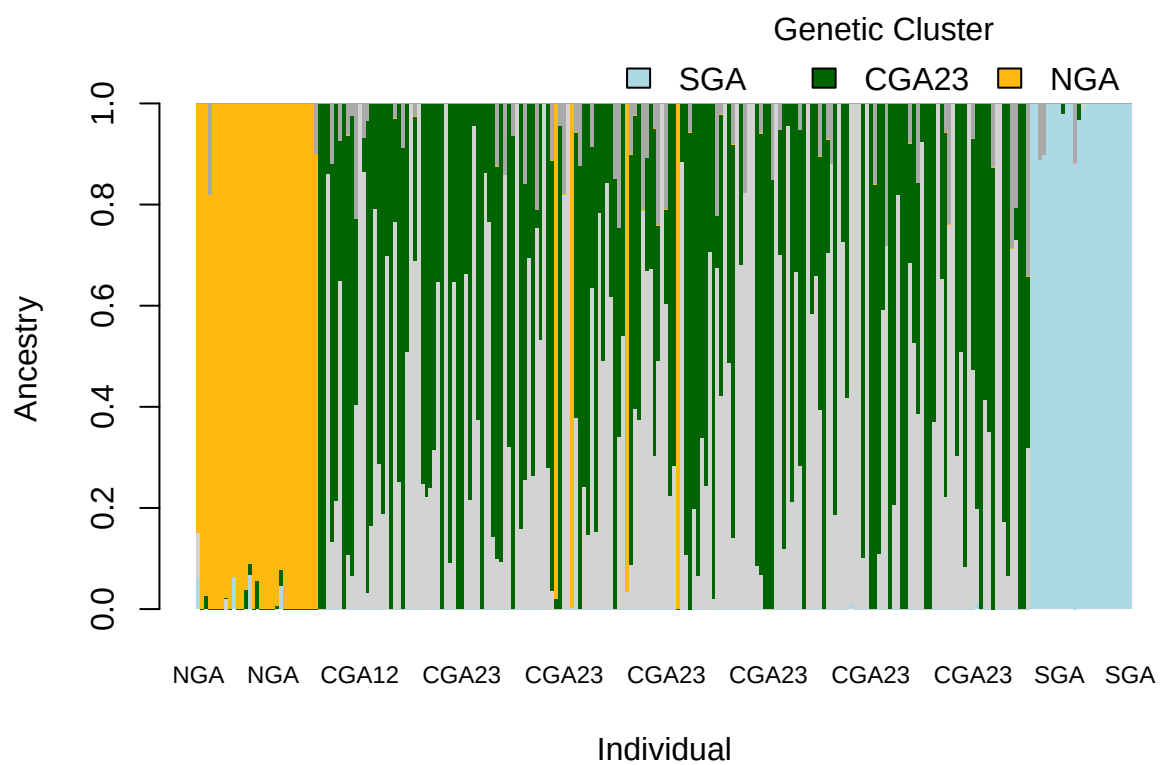
barplot(t(as.matrix(sortedQ[[k]])),
        col=colscheme[[k]], ylab = "Ancestry", xlab = "Individual",
        border=NA, names=sortedmap$pop, cex.names = 0.75)
legend("topright", bty="n", title = "Genetic Cluster",
       ncol = length(fillcols),
       fill = fillcols, legend = filllleg,
       xpd = TRUE, inset = c(0, -0.2))
}

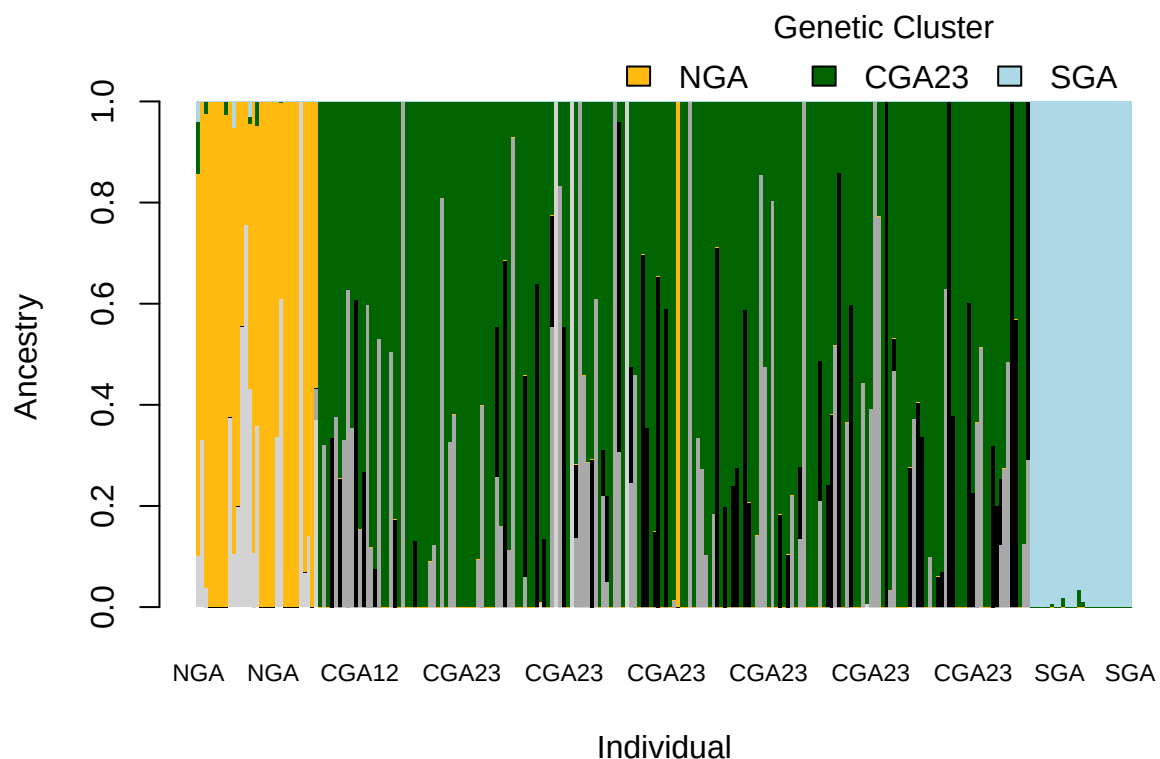
```











North GA bears in Central GA?

There are four bears in the central GA sample have north GA ancestry (four golden bars in the predominantly green sample set). Let's take a look at these bears...

```
#These show up with K = 2
colscheme[[2]]
```

```
## [1] "darkgreen"      "darkgoldenrod1"
```

```
nga.col <- which(colscheme[[2]] == popXcol$col[which(popXcol$pop == "NGA")])
sortedmap[which(sortedQ[[2]][,nga.col] > 0.5 &
  !sortedmap$pop %in% c("NGA", "SGA")),]
```

```
##                vcfID    pop sortord
## 61 UGA_173101_P015_WB09 CGA23        3
## 65 UGA_173101_P018_WF02 CGA23        3
## 79 UGA_173101_P015_WC07 CGA23        3
## 92 UGA_173101_P016_WE09 CGA23        3
```

```
samps[which(samps$vcfID %in%
  sortedmap$vcfID[which(sortedQ[[2]][,nga.col] > 0.5 &
    !sortedmap$pop %in% c("SGA", "NGA"))]),]
```

```

##               extID ugaID is.SCR is.rep week year snare utmSnare
## 21  UGA_173101_P015_WB09 74-B FALSE FALSE <NA> 2023 <NA> <NA>
## 31  UGA_173101_P015_WC07 73-B FALSE FALSE <NA> 2023 <NA> <NA>
## 153 UGA_173101_P016_WE09 83-RK FALSE FALSE <NA> 2023 <NA> <NA>
## 350 UGA_173101_P018_WF02 89-RK FALSE FALSE <NA> 2024 <NA> <NA>
##
##                                     fqfile
## 21  RAPiD-Genomics_F534_UGA_173101_P015_WB09_i5-502_i7-63_S14759_L003_R1_001.fastq
## 31  RAPiD-Genomics_F534_UGA_173101_P015_WC07_i5-502_i7-62_S14769_L003_R1_001.fastq
## 153 RAPiD-Genomics_F534_UGA_173101_P016_WE09_i5-504_i7-153_S14890_L003_R1_001.fastq
## 350 RAPiD-Genomics_F534_UGA_173101_P018_WF02_i5-537_i7-158_S15089_L003_R1_001.fastq
##      knownSEX sampSOURCE region nSNPgts          vcfID  FILT GTservice
## 21          M  Den Check    CGA   1234 UGA_173101_P015_WB09 FALSE      LGC
## 31          F  Den Check    CGA   1126 UGA_173101_P015_WC07 FALSE      LGC
## 153         M  Roadkill    CGA   1061 UGA_173101_P016_WE09 FALSE      LGC
## 350         M  Roadkill    CGA   1244 UGA_173101_P018_WF02 FALSE      LGC
##      sexGT COLclone finalClone finalSEX
## 21      M      86      86      M
## 31      F     136     136      F
## 153     M     155     155      M
## 350     M      68      68      M

```

Those are the same four bears we saw in the PCA. Offspring of a translocated NGA sow that mated with a male bear in the vicinity of Greene and Hancock counties (north of CGA).

Summary

Overall, the **admixture** analysis supports a solution with $K = 3$. The clustering at this K cleanly splits the three populations into separate clusters. Four bears with NGA ancestry are present in our CGA sample, but three of the four are offspring from a NGA sow that was translocated to CGA. The male she mated with in Hancock or Greene County (north of the CGA study region) must also have been of NGA origin (given the lack of evidence for CGA contributions to these samples). All three of these cubs were subsequently killed on the road by motorists, although one cub did make it back to CGA and appeared once in our SCR dataset. The other bear of NGA ancestry was also hit on the road near Augusta (again outside the CGA study region). Thus, the analyses here reveal no evidence for successful gene flow into the CGA population.

Overall, results from F_{ST} , PCA, and **admixture** are quite similar and in agreement with previous findings (*e.g.*, from [Hooker et al. 2019](#)). Back bears in central GA are more genetically differentiated from bears in north and south GA than the north and south GA populations are from one another (despite these populations being described as *Ursus americanus americanus* and *U. a. floridanus*, respectively). This is most likely due to strong genetic drift in the small central GA population, rather than a long history of genetic distinctiveness.
